

Analysis of Library Management System Transactions

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Stadtbibliothek

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Research Questions

- RQ 1: Spatial and Structural Analysis
  - How are library transactions distributed across different **locations and branches**?
  - Which libraries play a central role in the network of **inter-branch interactions**?
- RQ 2: Temporal Usage Pattern Analysis
  - How does library usage vary over **time(monthly, weekly, and daily)**?
- RQ 3: Genre and Demographic Analysis
  - How are transactions distributed across **different genres**?
  - How does genre preferences vary across **ages and genders**?

Data Overview

Time: Q4 in 2025 (2025-10-01 00:15:00 – 2025-12-31 23:40:00)

Type	Owning	Lending	Case	Format	Age	Gender	Genre
A	Motorama	Motorama	Own	Print	19	Male	Lifestyle
R	Giesing	Giesing	Own	DVD & Bluray	57	Female	Humanities
C	Sendling	Motorama	Inter	CD	21	Male	Literature
A	Motorama	HP8	Inter	Print	42	Female	Children & Youth
T	Neuhausen	Neuhausen	Own	Print	11	Female	Children & Youth

Transaction Types

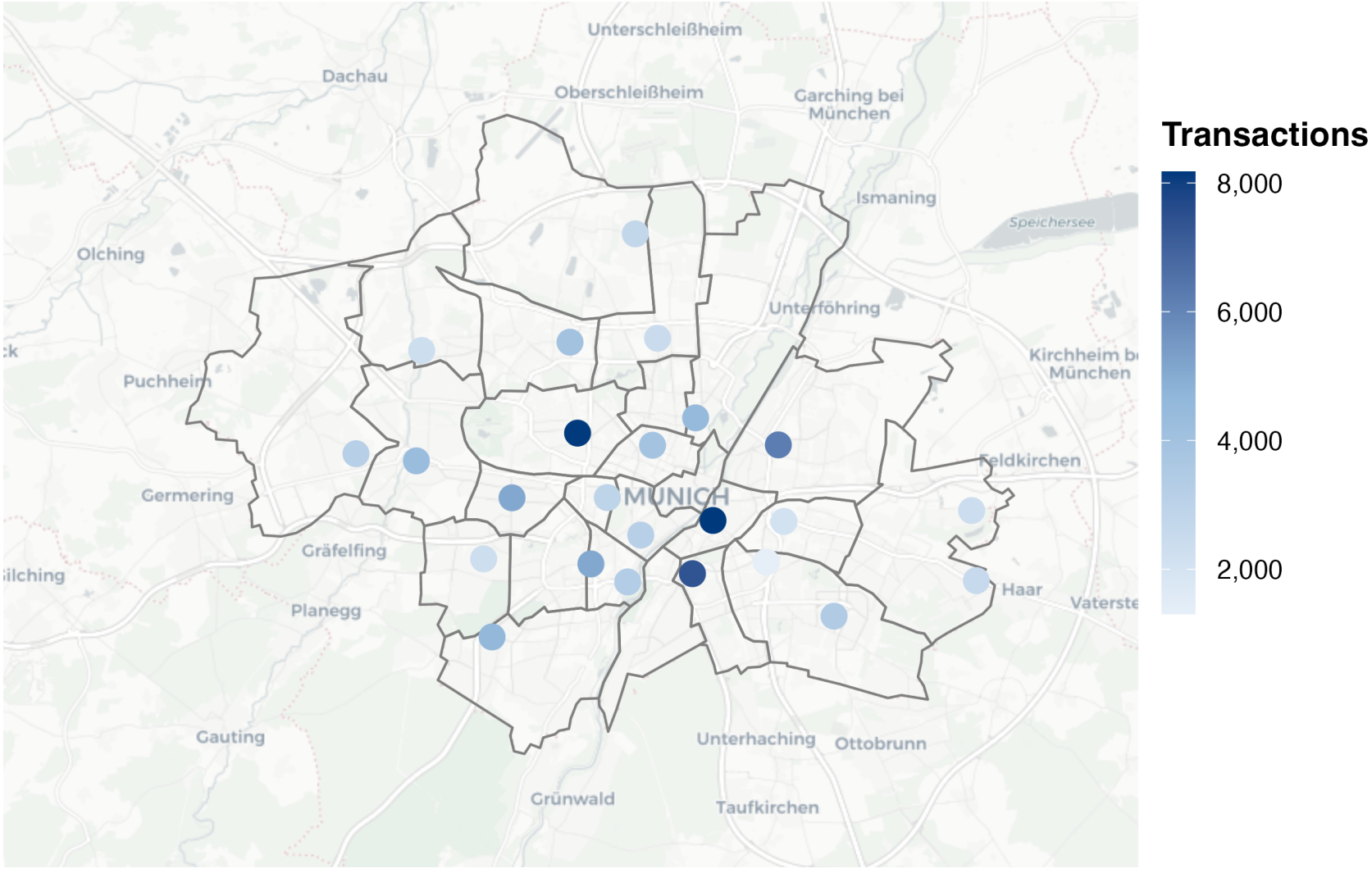
A = Borrow **R** = Return
T = Renewal **C** = Rejected renewal

- **RQ 1 & 3**: Borrow transactions
- **RQ 2**: Borrow & Return transactions

Data Size Before Cleaning: 4,689,173 rows
After Cleaning: 1,489,147 rows

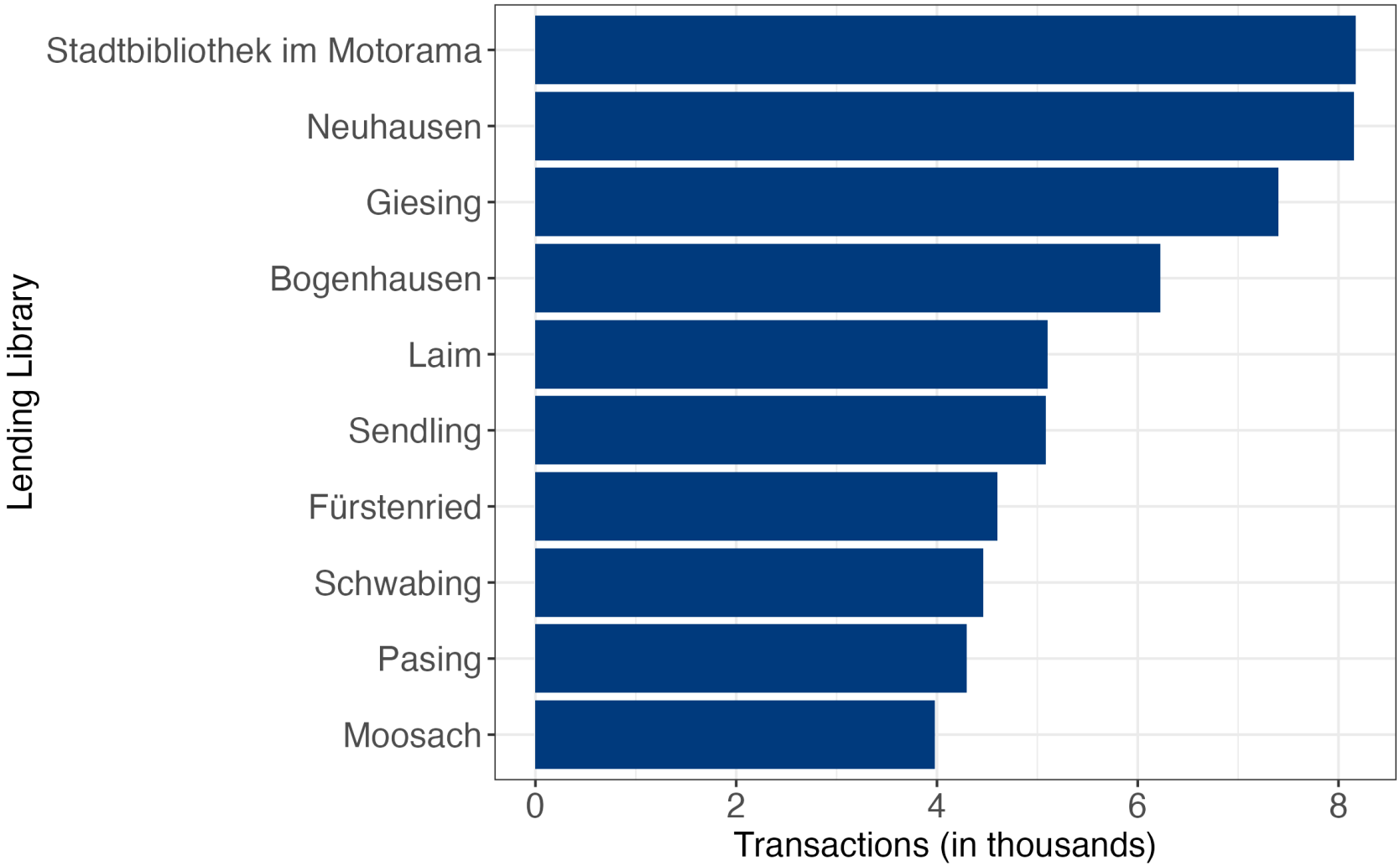
RQ1: Spatial and Structural Analysis

Library Usage Across Munich

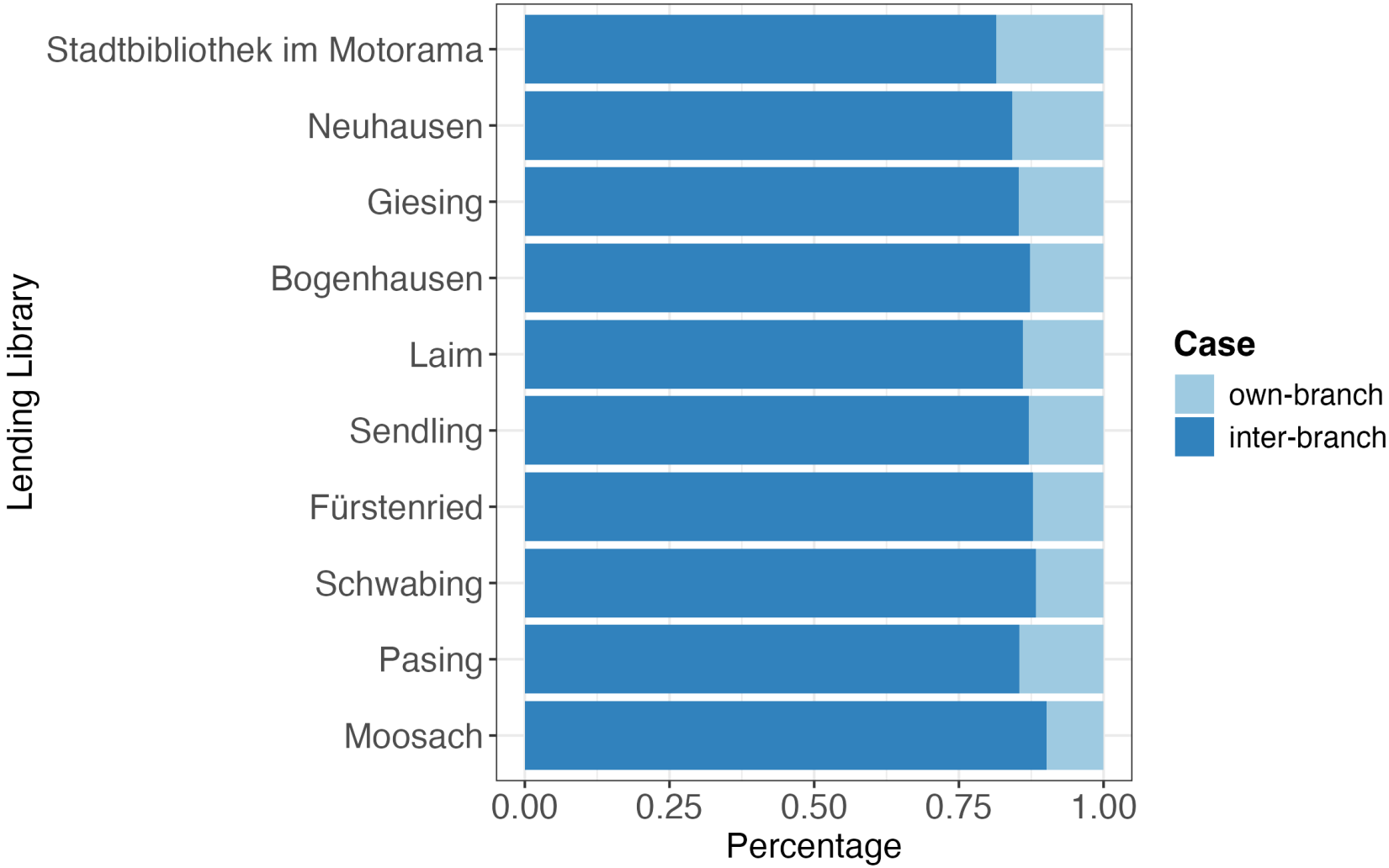


- First part of the analysis is based on **borrowing transactions** only (returns and invalid ID excluded)

Top Lending Libraries by Number of Transactions

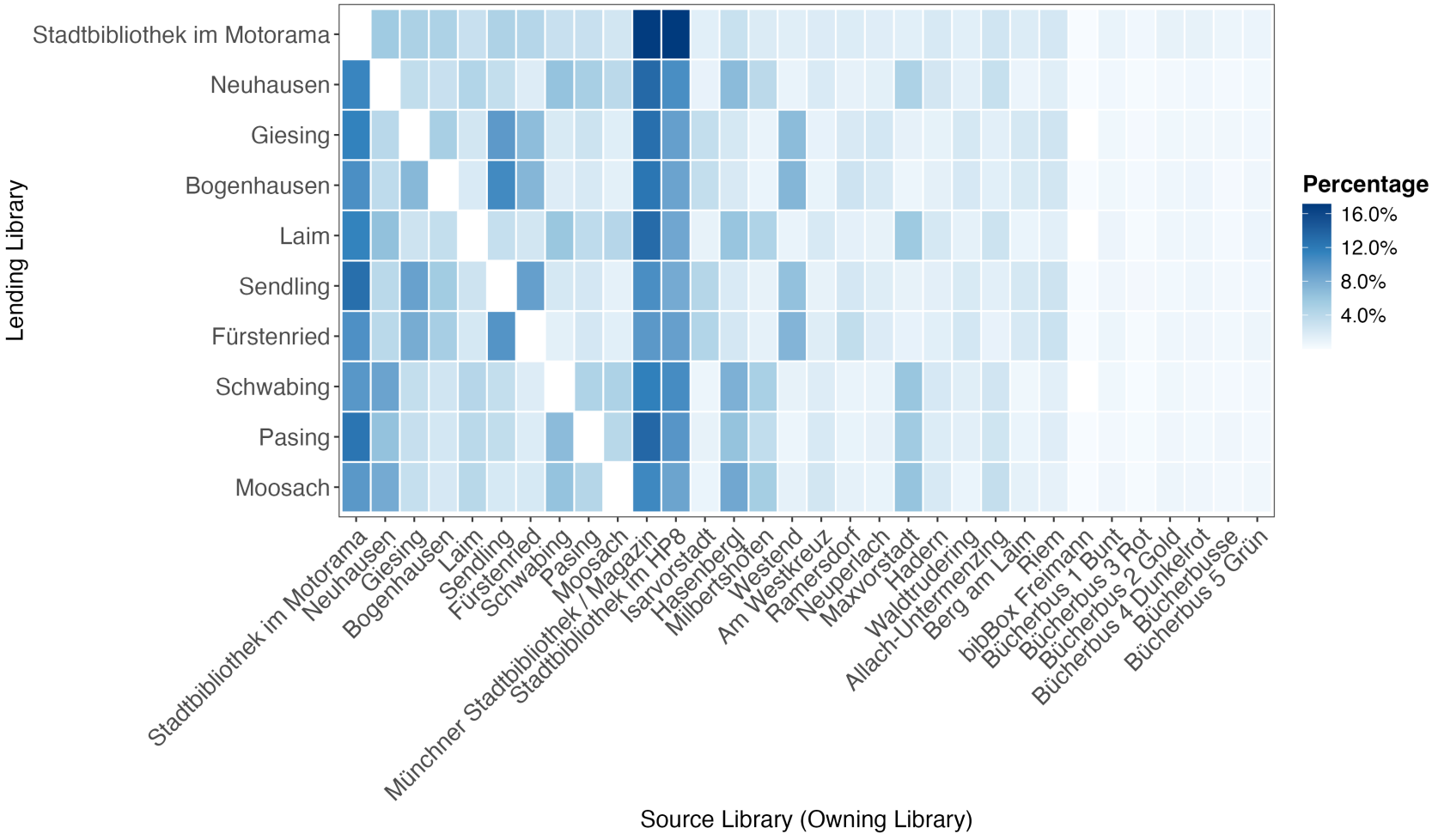


Own-Branch vs Inter-Branch Borrowing by Library



- On average, about 86% of inter-branch transactions

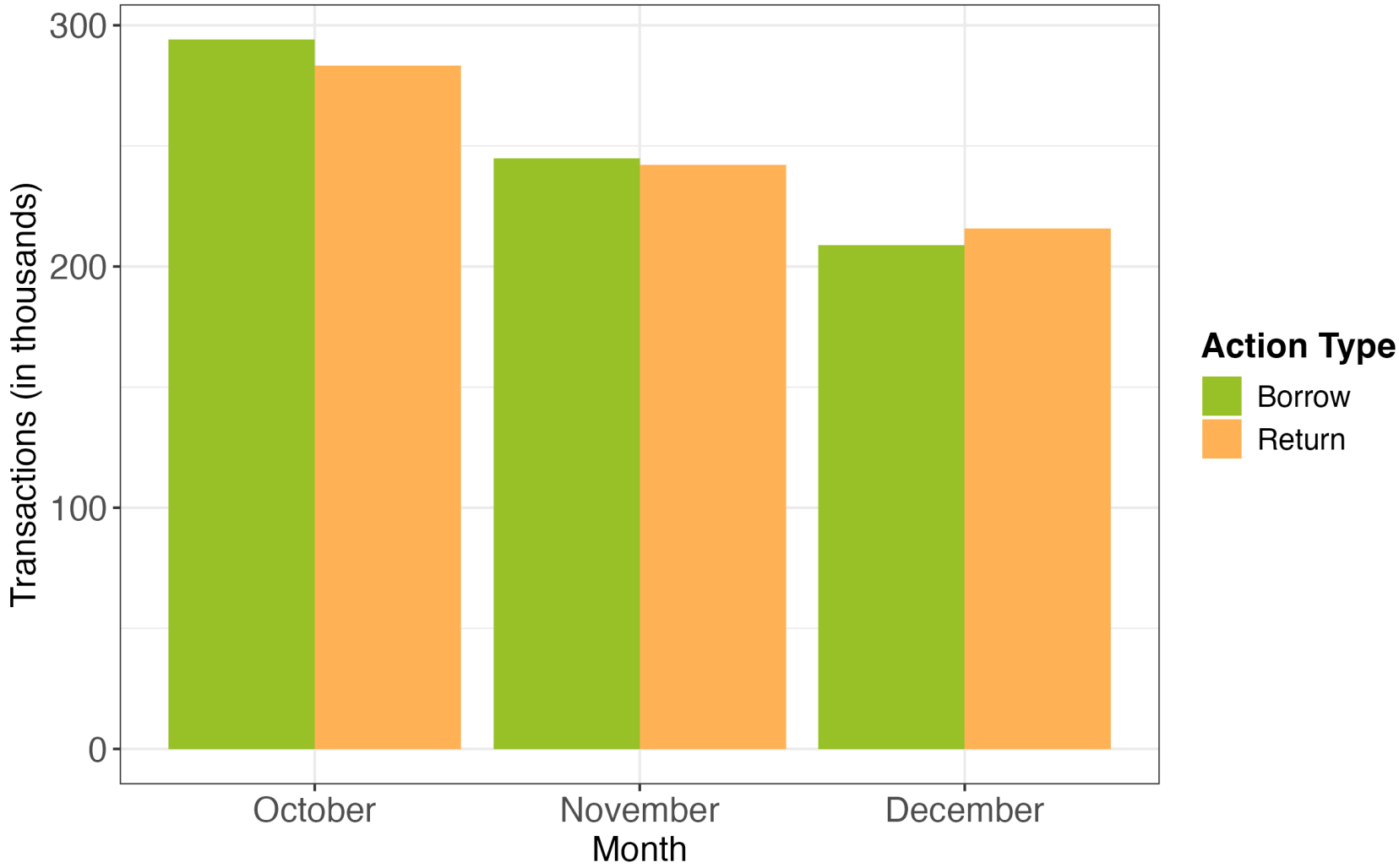
What Are the Sources of Inter-branch Transactions?



- On average, ~32% of inter-branch transactions (top 10 libraries) come from three major hubs

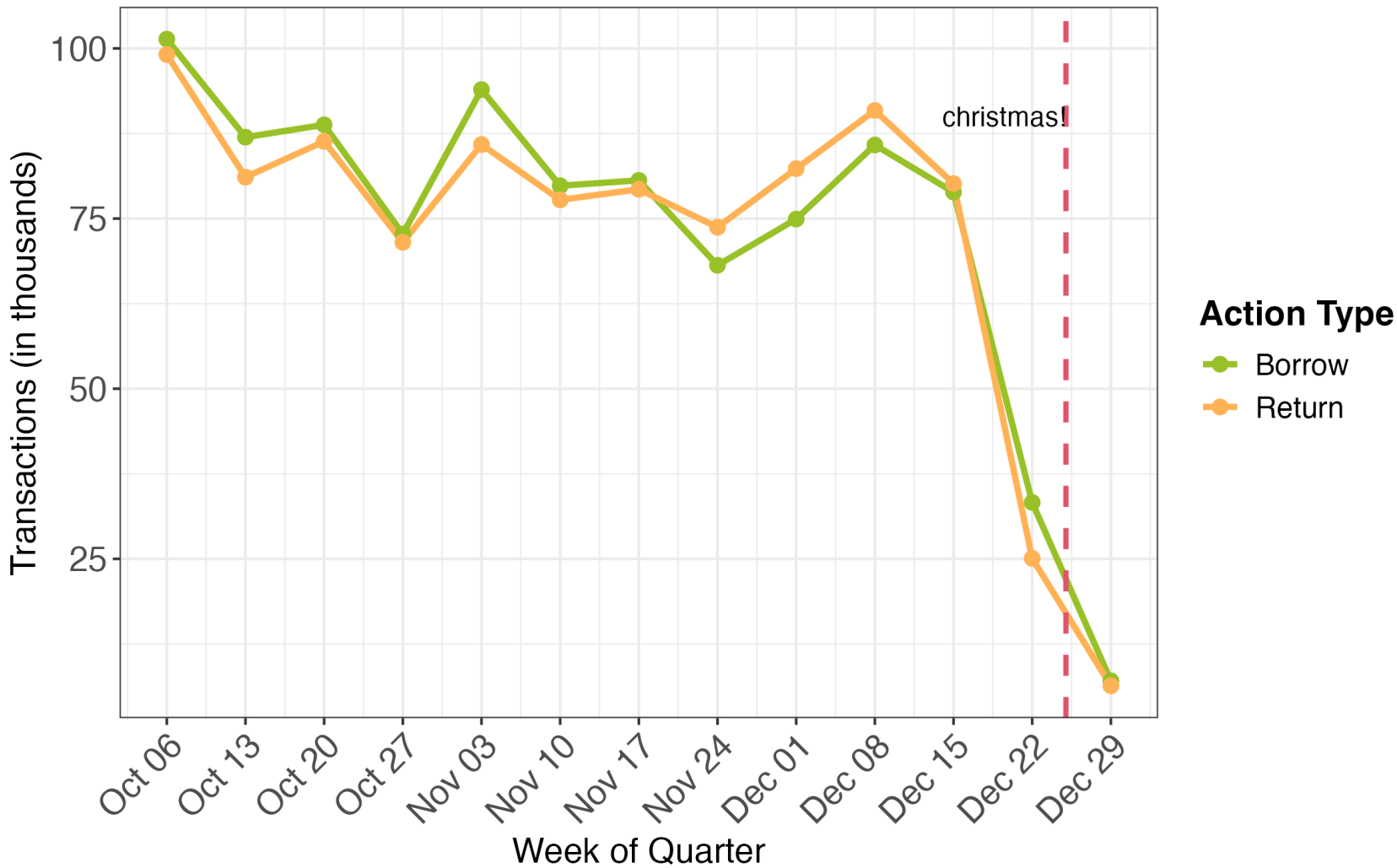
RQ2: Temporal Analysis

Monthly Library Transactions

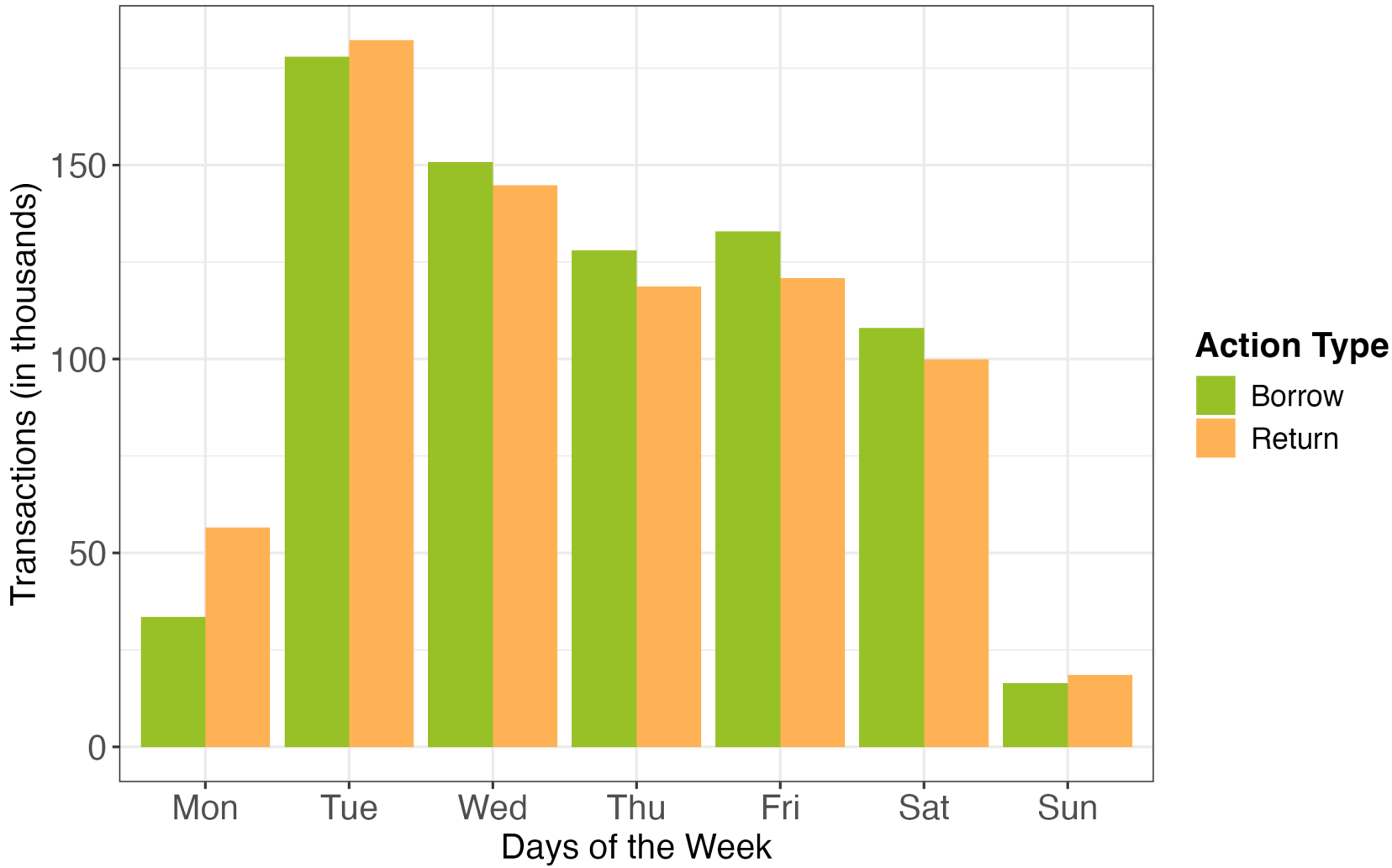


- Second part of the analysis is based on **borrowing transactions** and **returning transactions**.

Weekly Library Transactions

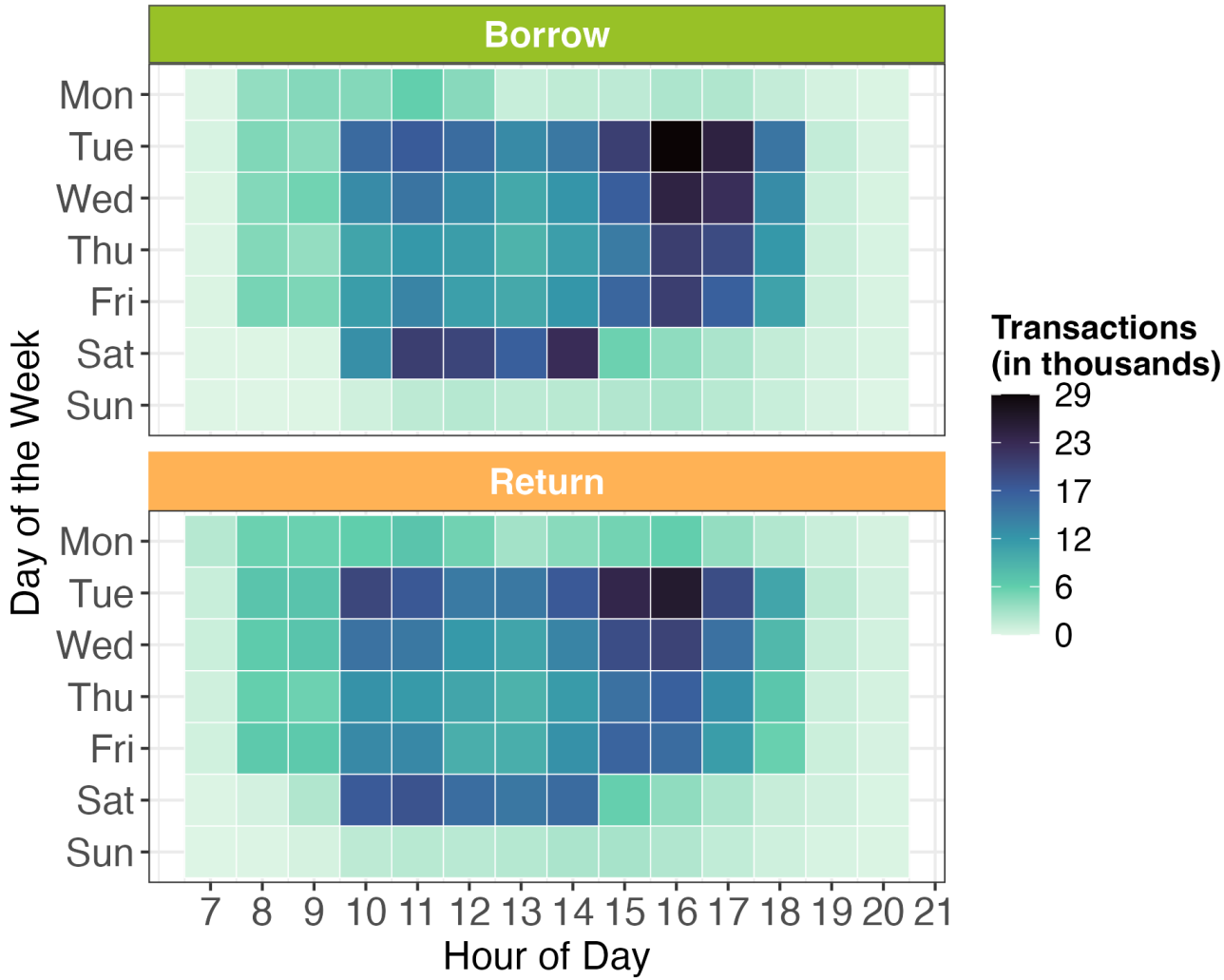


Transaction Pattern by Day of the Week



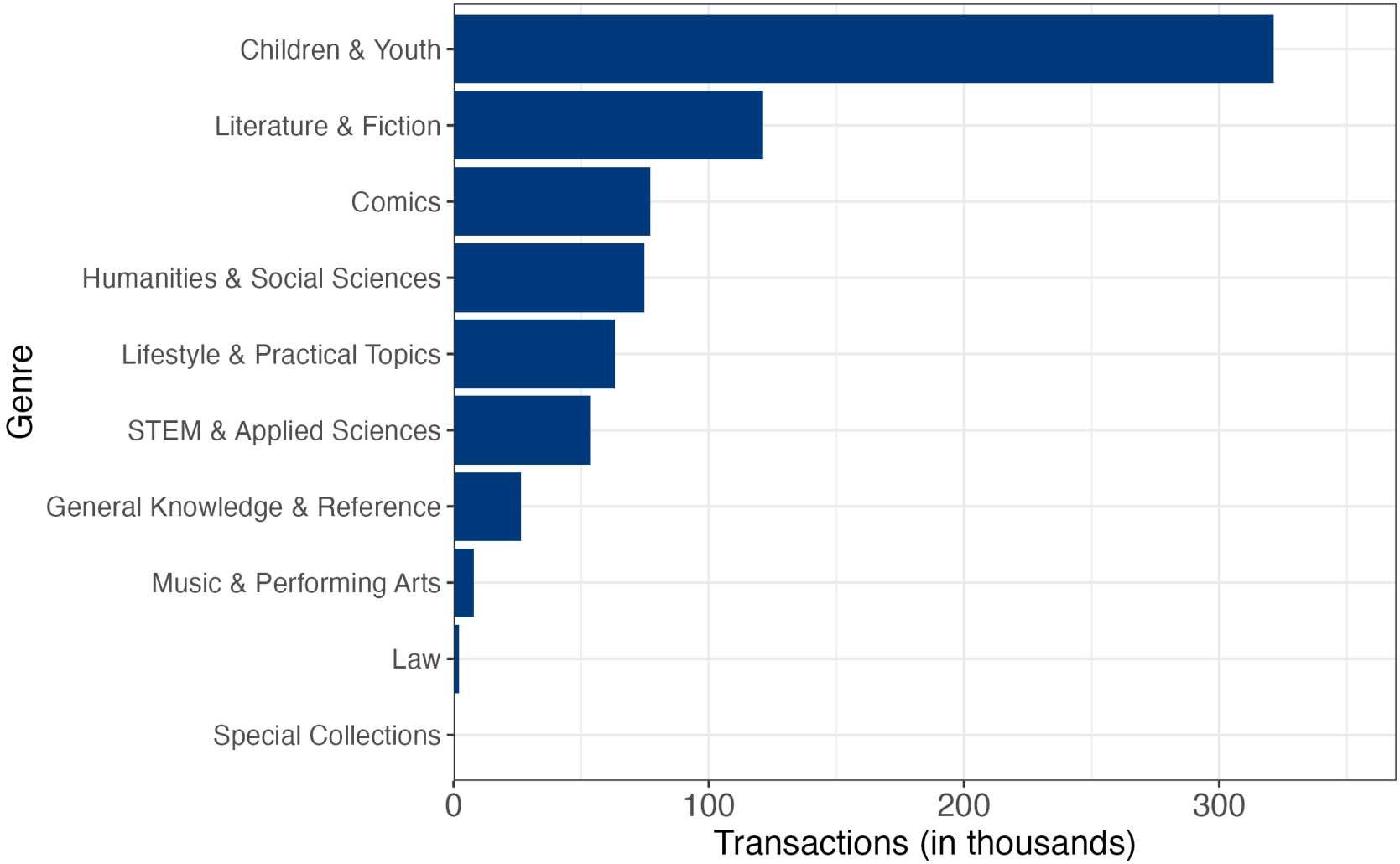
- Some libraries are close on Monday and Sunday.

Hourly Library Transactions by Day of the Week



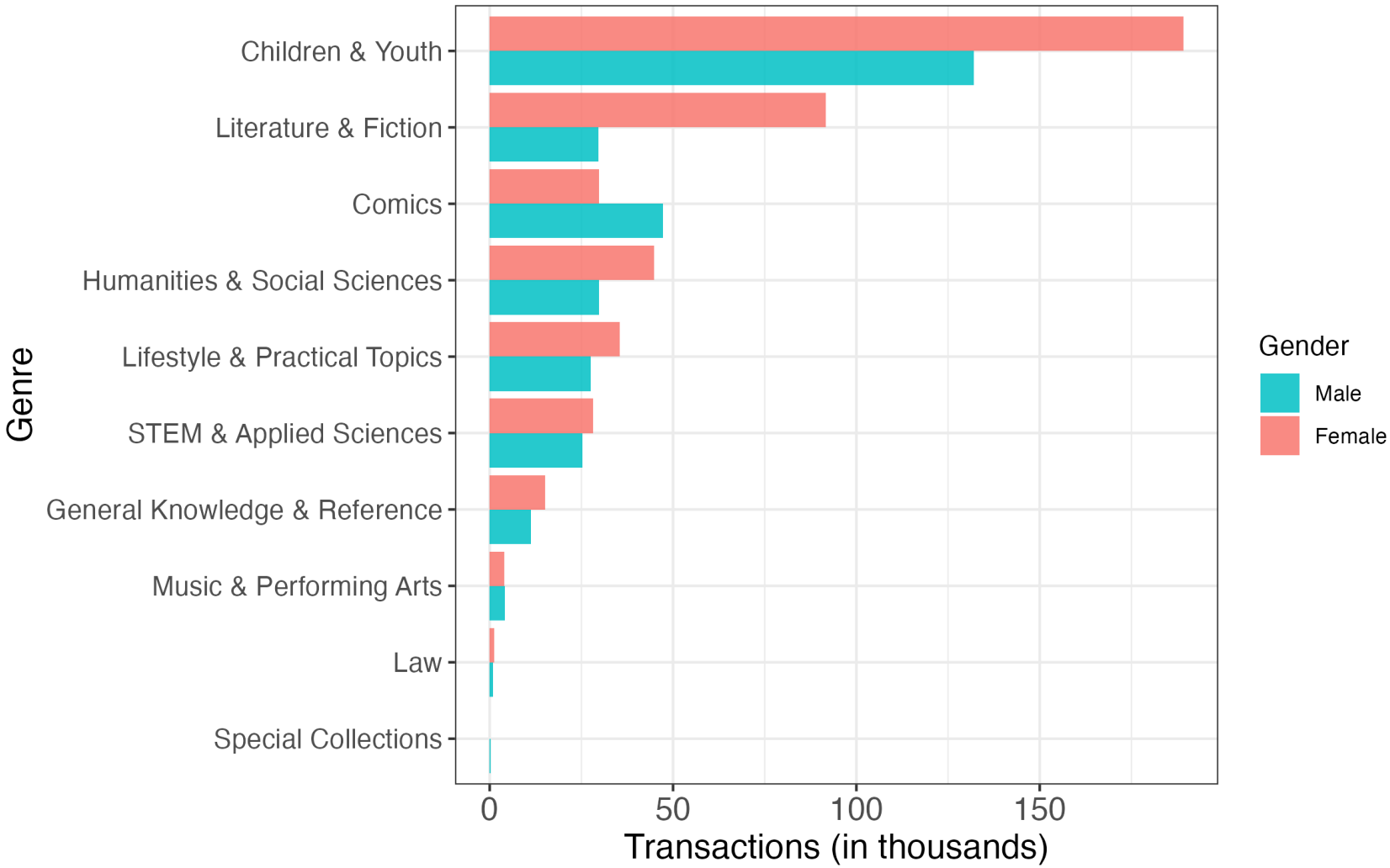
RQ3: User Preference Analysis

Library Borrows by Genre

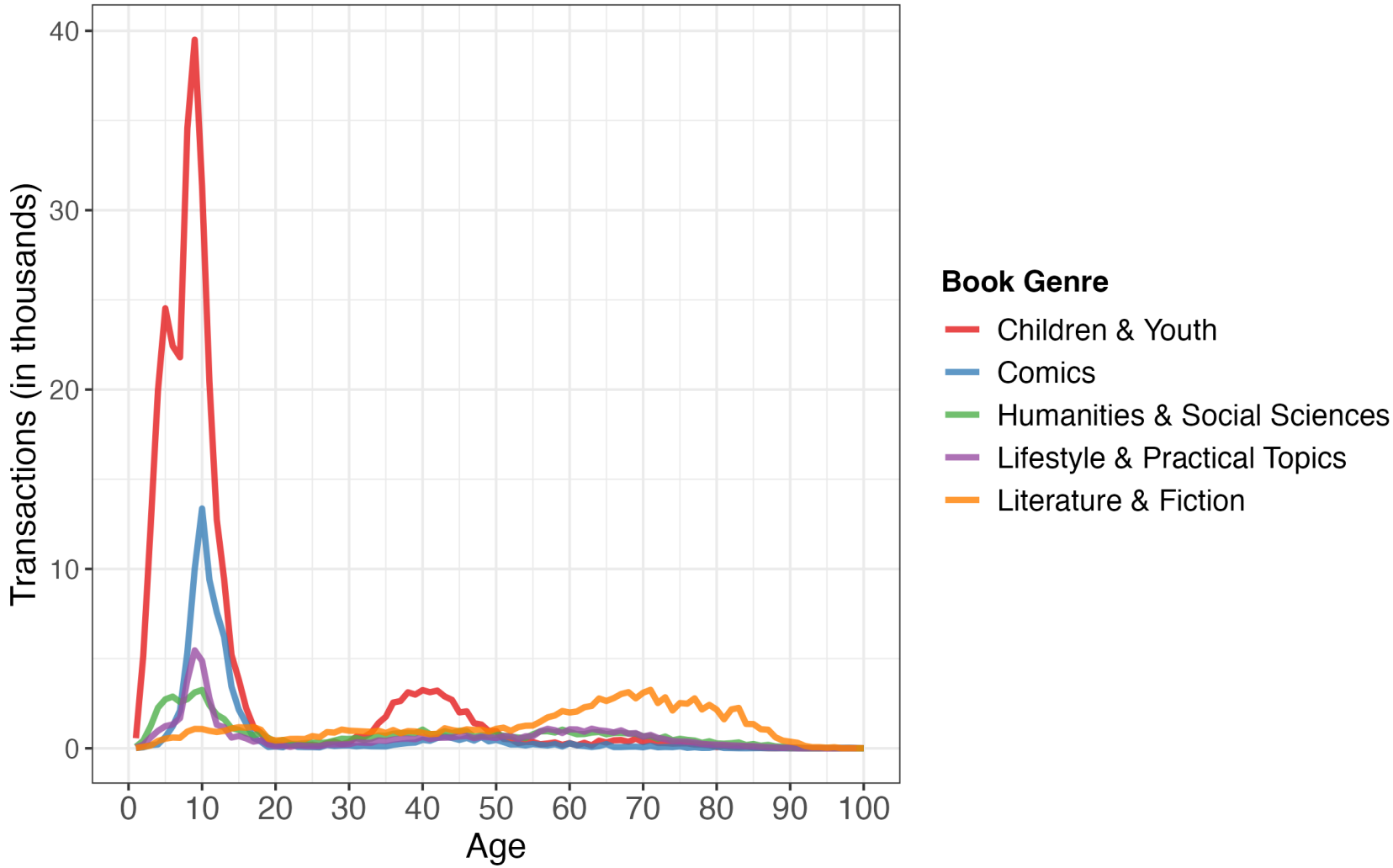


- Third part of the analysis is based on **borrowing transactions** only. (returns excluded)

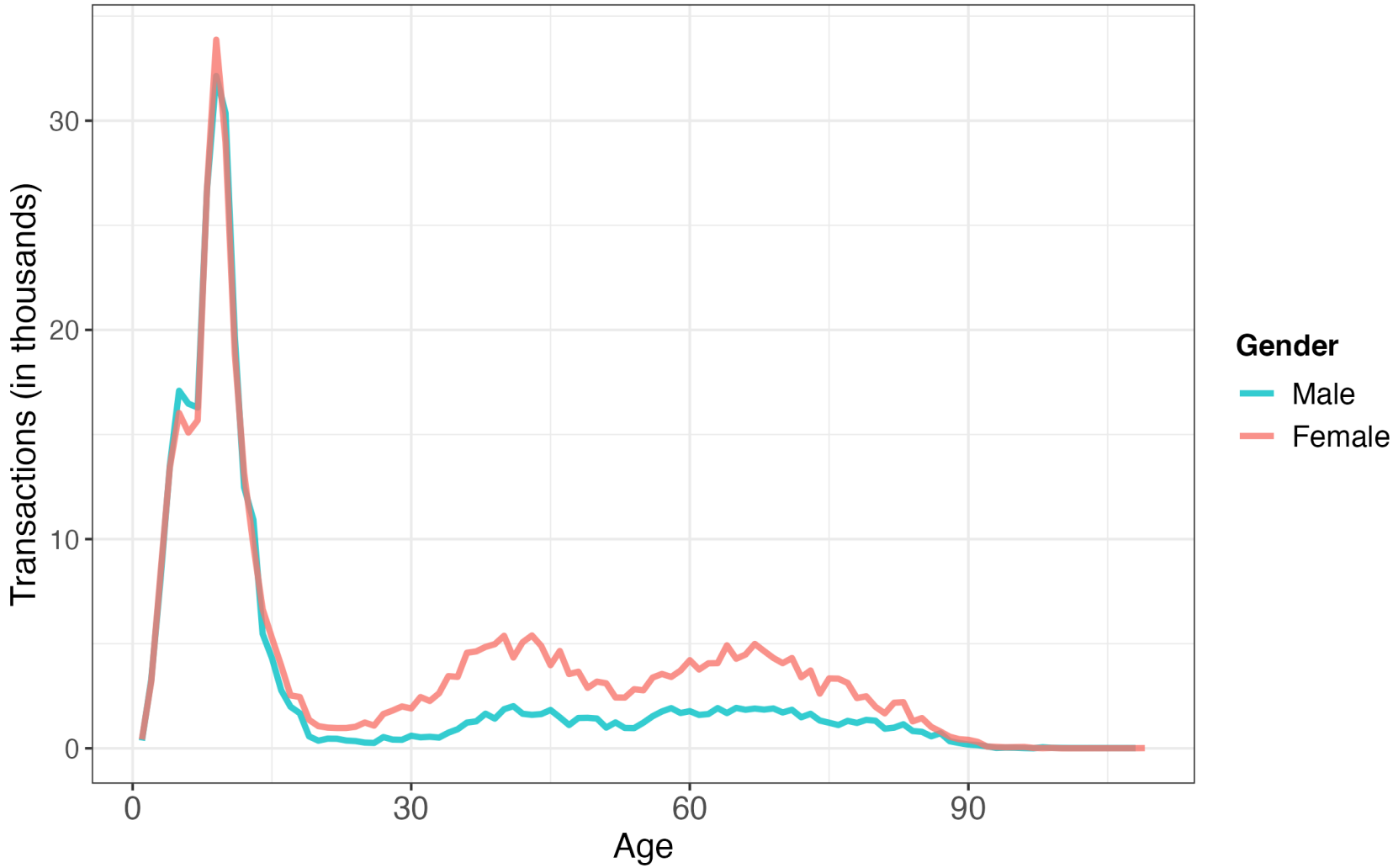
Genre Preference by Gender



Book Genre Preferences Across Age Groups



Borrowing Density by Age and Gender



Key Findings and Conclusion

- Libraries located in central areas are used more frequently.
- The Münchner Stadtbibliothek, HP8, and Motorama act as major hubs for interbranch transactions, but borrowing sources remain diverse.
- A noticeable decline in activity occurs during the Christmas season(After 15th of Dec).
- Tuesday showed the highest level of usage, especially between 3pm and 5pm.
- Users aged 7 to 12 represent the most active group (Children & Youth).
- Genre preferences vary across different age groups and genders.

Limitation

- The lending library ID contains many invalid values (-1).
  After removing these entries, only about 14% of the original data remained (~86% excluded).
  This mainly affects the spatial Analysis (Part 1), while other analyses are not impacted.
- The dataset only covers Q4 2025 (Oct –Dec).
  Seasonal patterns across other periods cannot be analyzed.
- Categories such as diverse and institute (~11%) were excluded.

Discussion

- A small number of central libraries (e.g., Motorama, HP8, Magazin) act as key hubs, but inter-branch borrowing remains widely distributed across the network
- Library usage follows clear temporal patterns, with strong weekday peaks (especially Tuesday afternoons) and a sharp decline during Christmas
- User behavior is highly demographic-driven: younger users (7–12) dominate activity, and genre preferences vary by age and gender

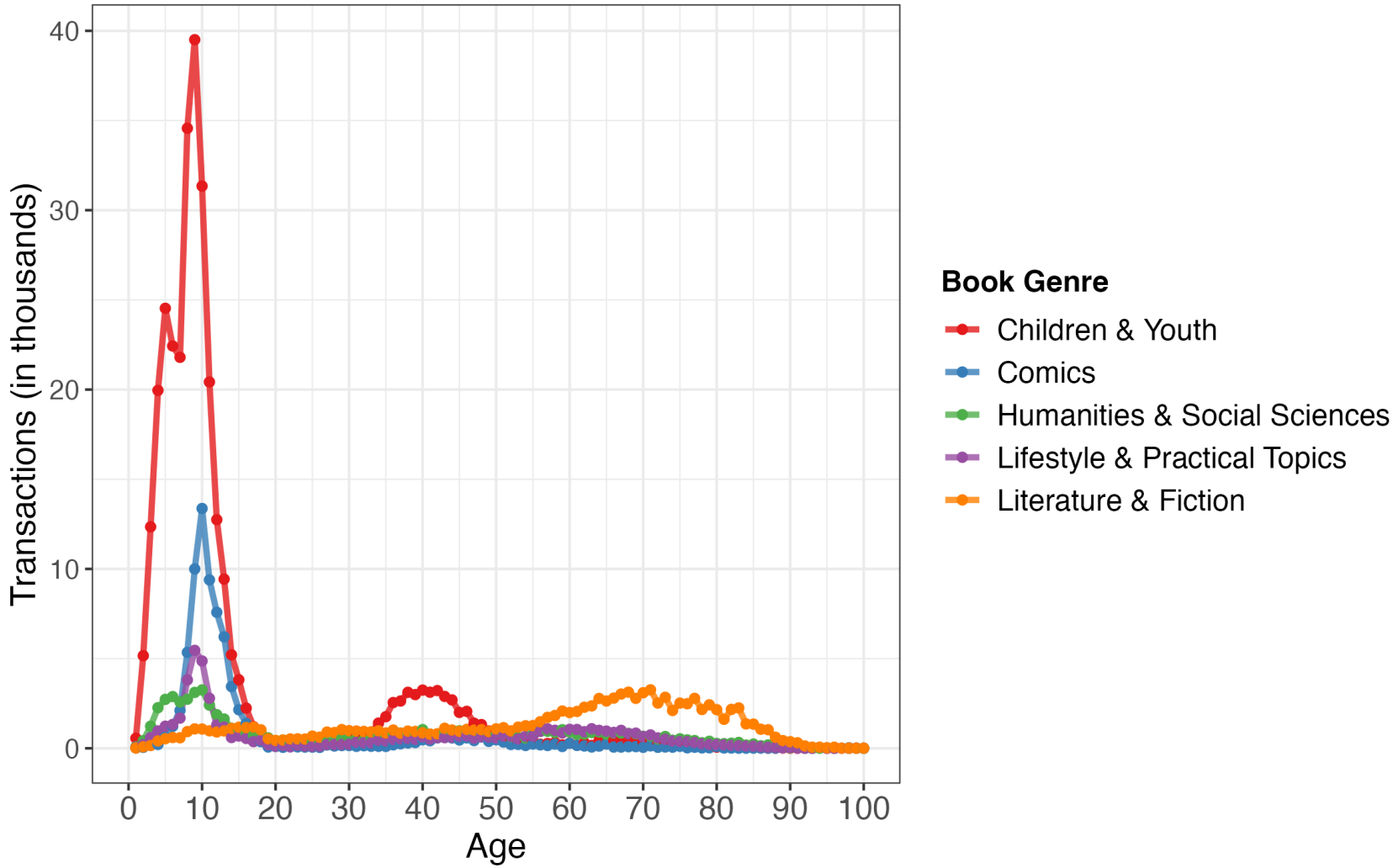
$\rightarrow$ How can these patterns be used to optimize resource allocation, staffing, and collection distribution across libraries?

Appendix

Why are there some late transaction?

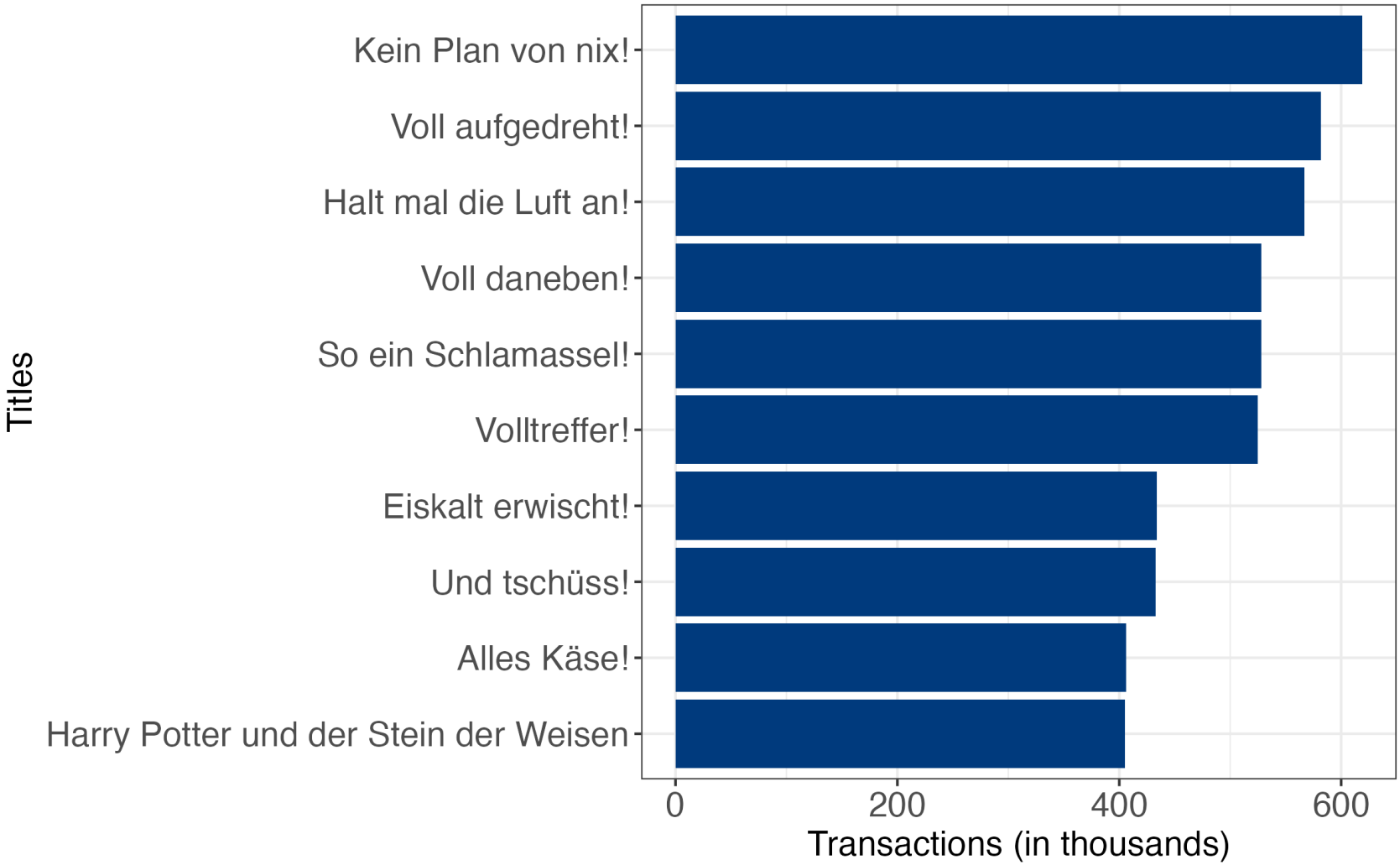
- **Borrowing / Pick-up**
  - Stadtbibliothek Giesing, Stadtbibliothek im Motorama open until 22.
  - Stadtbibliothek im HP8 opens until 23.
- **Returning**
  - Book returns are possible outside opening hours via return boxes; however, timestamps may reflect delayed processing rather than the exact time of return.

Why is the plot not jagging in the interval?



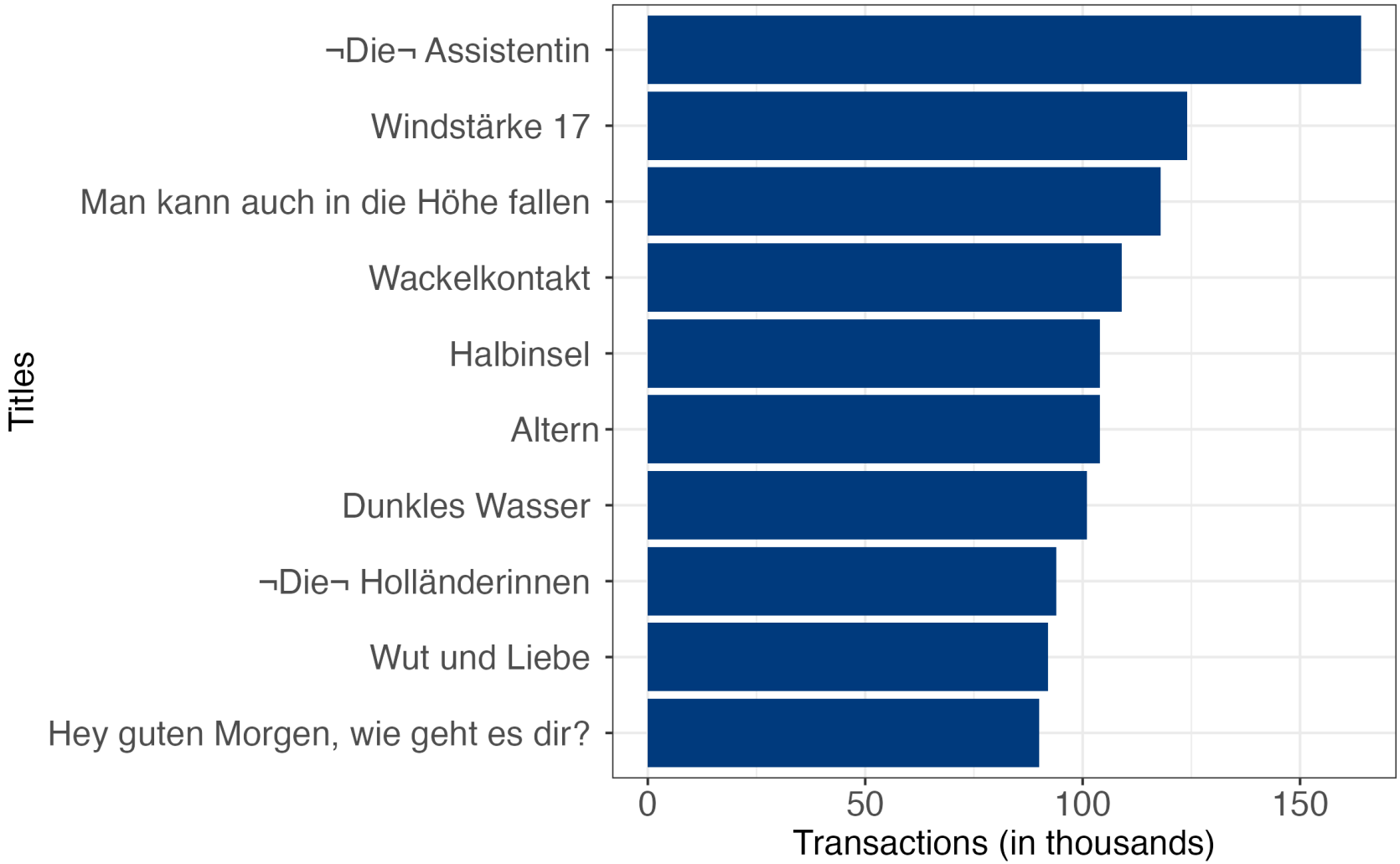
- Smoother patterns at younger ages result from more continuous transitions between adjacent age groups, not differences in data availability.

Top10 Book Titles among Children & Youth



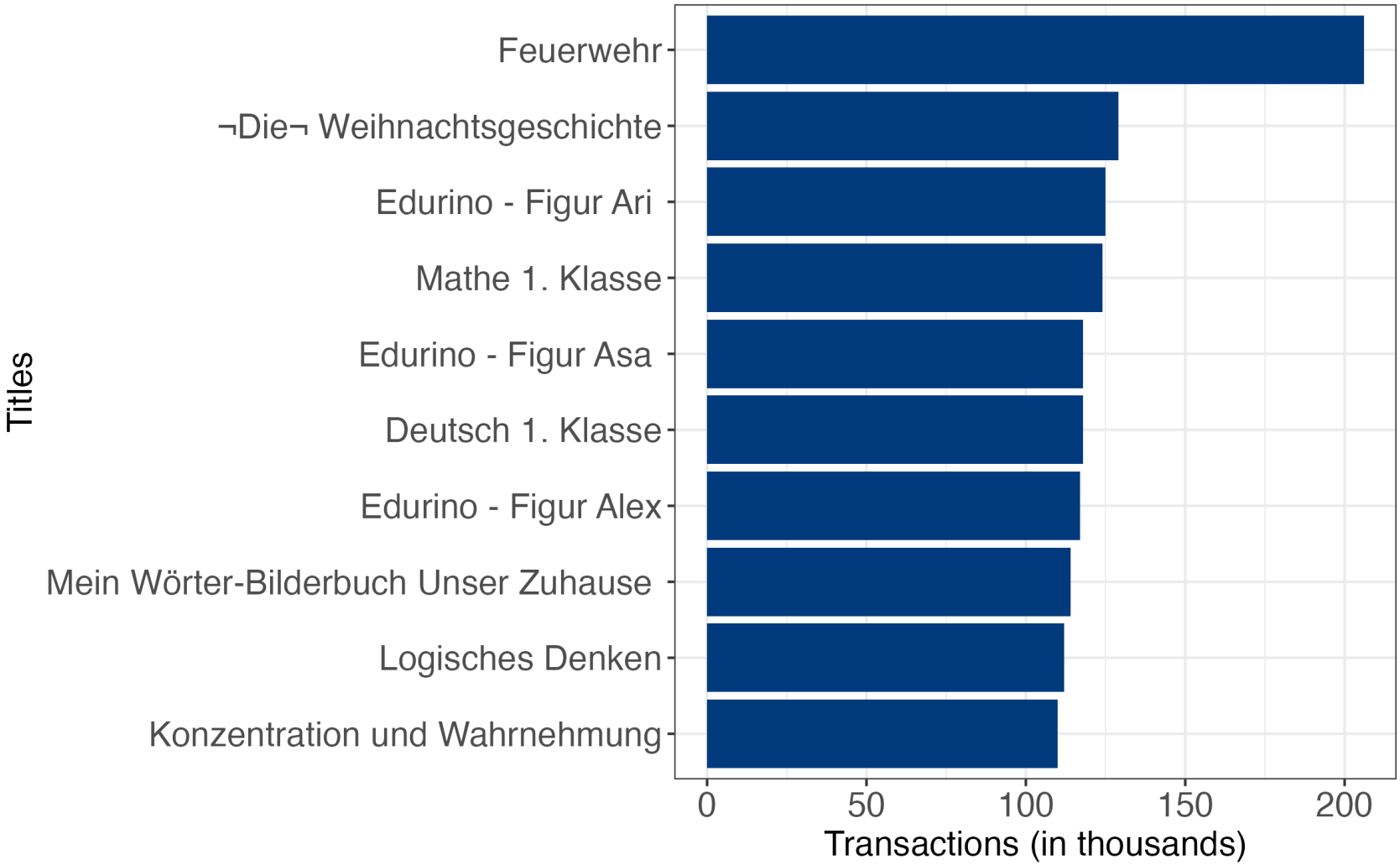
- Strong preference for humorous and easy-to-read books with playful, everyday language

Top10 Book Titles among Literature & Fiction



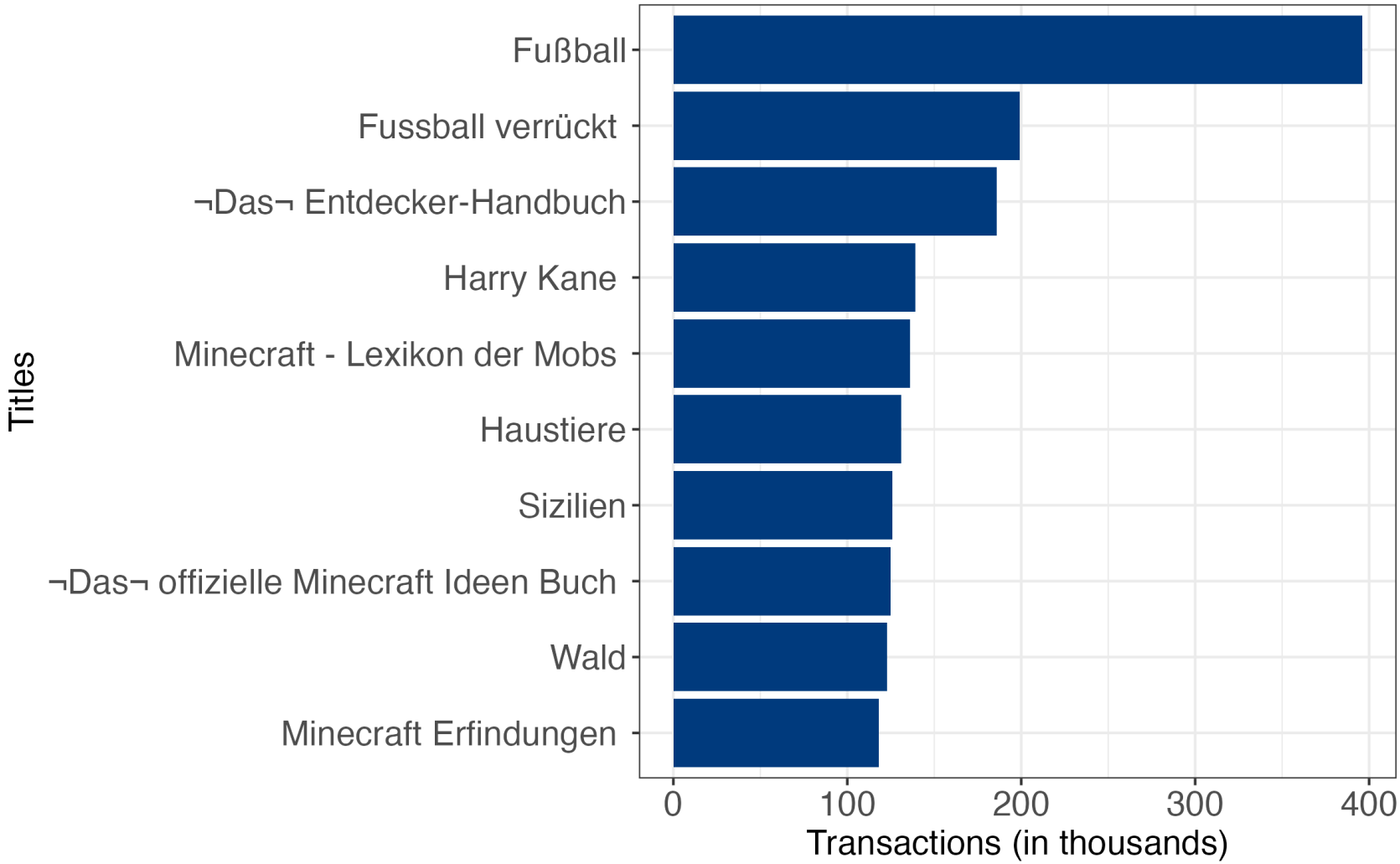
- Titles suggest a focus on adult-oriented themes such as relationships, emotions, and personal development

Top10 Book Titles among Humanities & Social Sciences



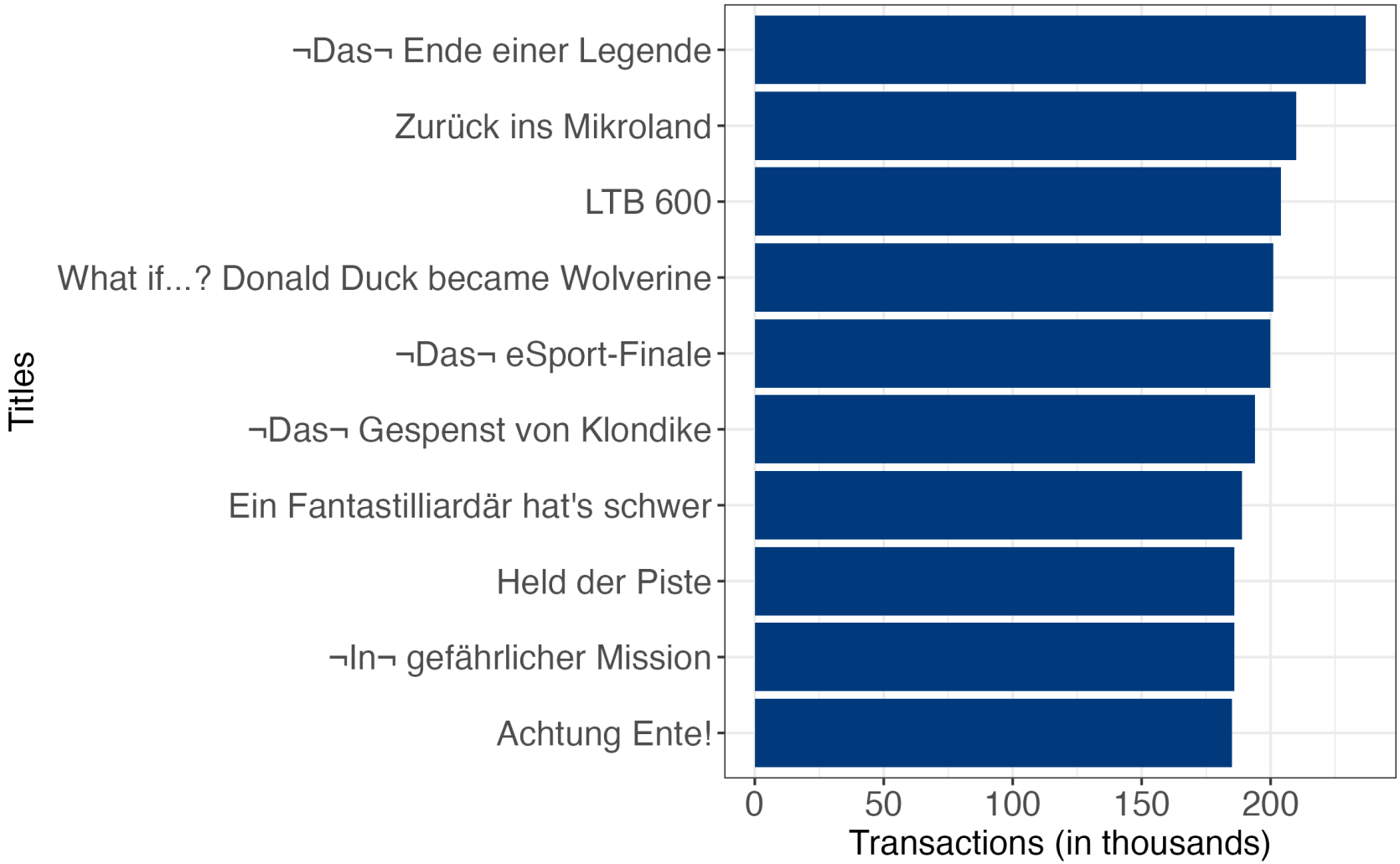
- Some hybrid materials (e.g., Edurino) are classified as PRINT and thus included. Top titles are primarily educational, focusing on school learning and cognitive skill development.

Top10 Book Titles among Lifestyle & Practical Topics



- Borrowing is driven by interest-based topics, particularly sports and gaming.

Top10 Book Titles among Comics



- All top titles belong to the "Walt Disney Lustiges Taschenbuch" series, indicating a strong dominance of a single comic franchise.

Full Heatmap (add row-wise up to 100%)

